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Course Code:DSA0502
Couse Name:QUERY PROCESSING FOR DATA SCIENCE

CO1 AT2 Application-Based Questions

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9. Weather Data Processing
Weather data is stored in weather.csv.
Task:
Read the file.
Find the highest and lowest temperatures.
Calculate average rainfall.
Display days with rainfall greater than 100 mm.

Aim

To read weather data from a CSV file, find the highest and lowest temperatures, calculate the average rainfall, and display days with rainfall greater than 100 mm using Pandas.

Procedure

1. Import the Pandas library.
2. Read the CSV file using `read_csv()`.
3. Find the highest temperature using `max()`.
4. Find the lowest temperature using `min()`.
5. Calculate average rainfall using `mean()`.
6. Filter rainfall greater than 100 mm.
7. Display the results.

```
import pandas as pd



# Read the weather CSV file

df = pd.read_csv("weather.csv")
```

```python
# Display the weather data

print("Weather Data:")

print(df)


# Find highest temperature

highest = df["Temperature"].max()


# Find lowest temperature

lowest = df["Temperature"].min()


# Calculate average rainfall

average_rainfall = df["Rainfall"].mean()


# Find days with rainfall greater than 100 mm

heavy_rain = df[df["Rainfall"] > 100]


# Display results

print("\nHighest Temperature:", highest)

print("Lowest Temperature:", lowest)

print("Average Rainfall:", average_rainfall, "mm")


print("\nDays with Rainfall Greater than 100 mm:")

print(heavy_rain)
```

```
PS C:\Users\sandr\OneDrive\Desktop\WeatherDataProcessing> python weather_processing.py
Weather Data:
         Day  Temperature  Rainfall
0     Monday           32        80
1    Tuesday           35       120
2  Wednesday           30        50
3   Thursday           38       150
4     Friday           33        90
5   Saturday           29       110
6     Sunday           36        70

Highest Temperature: 38
Lowest Temperature: 29
Average Rainfall: 95.71428571428571 mm

Days with Rainfall Greater than 100 mm:
        Day  Temperature  Rainfall
1   Tuesday           35       120
3  Thursday           38       150
5  Saturday           29       110
PS C:\Users\sandr\OneDrive\Desktop\WeatherDataProcessing> 
```

Result

The weather data was successfully analyzed. The highest and lowest temperatures, average rainfall, and days with rainfall greater than 100 mm were displayed.

10. University Admission Data
Student admission data is stored in admissions.csv.
Task:
Identify duplicate student records.
Remove duplicate entries.
Display students admitted to the CSE department.
Save the updated dataset.

Aim

To read weather data from a CSV file, find the highest and lowest temperatures, calculate the average rainfall, and display days with rainfall greater than 100 mm using Pandas.

Procedure

1. Import the Pandas library.
2. Read the CSV file using `read_csv()`.
3. Find the highest temperature using `max()`.
4. Find the lowest temperature using `min()`.
5. Calculate average rainfall using `mean()`.
6. Filter rainfall greater than 100 mm.
7. Display the results.

```
import pandas as pd



# Read the admission data

df = pd.read_csv("admissions.csv")



# Display original data

print("Original Data:")

print(df)



# Find duplicate records
```

```python
duplicates = df[df.duplicated()]


print("\nDuplicate Records:")

print(duplicates)


# Remove duplicate records

df = df.drop_duplicates()


# Display students admitted to CSE

cse_students = df[df["Department"] == "CSE"]


print("\nStudents Admitted to CSE:")

print(cse_students)


# Save updated dataset

df.to_csv("updated_admissions.csv", index=False)


print("\nUpdated dataset saved as updated_admissions.csv")
```

```
PS C:\Users\sandr\OneDrive\Desktop\UniversityAdmission> python admission_processing.py
Original Data:
   StudentID     Name Department
0        101    Alice        CSE
1        102      Bob        ECE
2        103  Charlie        CSE
3        104    David        EEE
4        102      Bob        ECE
5        105      Eva        CSE
6        103  Charlie        CSE

Duplicate Records:
   StudentID     Name Department
4        102      Bob        ECE
6        103  Charlie        CSE

Students Admitted to CSE:
   StudentID     Name Department
0        101    Alice        CSE
2        103  Charlie        CSE
5        105      Eva        CSE

Updated dataset saved as updated_admissions.csv
PS C:\Users\sandr\OneDrive\Desktop\UniversityAdmission>
```

Result

The weather data was successfully analyzed. The highest and lowest temperatures, average rainfall, and days with rainfall greater than 100 mm were displayed.